



Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division

Course: AIML ZG528 – AI and ML for Robotics
Assignment 1: Modeling and Evaluating Robot Motion and Perception in Workplace Contexts

Modules Covered Module 1 – Introduction to Robotics;

Module 2 – Components of Programmable Robotic System

Total Marks:10

Deadline: As announced on eLearn

Type: Work-Integrated Learning Assignment (Part A - Group , Part B -Individual)

Submission: PDF report (3–4 pages) + Python code notebook with screenshots

Learning Objectives

After completing this assignment, students will be able to:

- Apply robotics and motion modeling concepts to realistic workplace-inspired problems.
- Evaluate how robot components (sensors, actuators, motion models) perform in specific work environments.
- Reflect on how robotics concepts connect to their own professional or industrial context.

Part A – Application (7 Marks)

Scenario: You are working in a team deploying autonomous robots in a workplace setting. **Choose any one domain based on your group's (team's) preference from below. Though possible horizontal areas are listed below, feel free to use your imagination:)**

- Inventory bots (movement, management, monitoring etc.,)
- Warehouse bots (logistics, inspection etc.,)
- Healthcare bots (delivery, monitoring, assistance etc.,)
- Hospitality bots (delivery, monitoring, assistance etc.,)
- Manufacturing bots (process inspection, material handling, assembly etc.,)
- Agricultural bots (inspection, material handling, delivery, harvesting etc.,)
- Environmental field bots (monitoring, inspection, deployment, rescue, clearing, sample collection etc.,)
- Aviation bots (assistance, search, combat, logistics etc.,)
- Software bots (any automation agents where all the phases of robotics must be applicable)

1. Robot System Description (2 Marks)

- a) Describe the type of robot, its function, and working environment
- b) Identify two proprioceptive and two exteroceptive sensors suitable for this robot and justify your choices.

2. Motion Model Implementation (4 Marks)

- a) Implement a Python program to illustrate the application of probabilistic motion models for the selected domain & your group's problem statement design scope.
 - i) For motion model : Use either velocity or odometry-based designs only



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- ii) For sensor model : Use beam model based designs only
- b) Demonstrate how uncertainty affects motion (e.g., using random noise or probability functions). Include a short visualization in the python code for above part a) implementation (trajectory plot or simulated path).

3. Workplace Evaluation (1 Marks)

Discuss how the chosen model in above Part 1) & implemented design in Part 2) and components would impact performance, safety, and productivity in your workplace context. Illustrate in no more than 5 well-articulated sentences.

Part B – Individual Reflection (3 Marks)

Students must submit their own reflection, even if the simulation or analysis is done collaboratively. [Click on this link to access the form \(only via BITS email ID\) to individually enter your responses to below questions.](#)

In the form above you would write a reflection addressing the following broad areas:

1. Workplace Connection (1 Marks) – How does this assignment relate to your current or aspirational workplace? Which robotics concept(s) did you find directly applicable to your field?
2. Learning Reflection (1 Marks) – What practical insights did you gain about robot motion or sensing from the implementation? What was the most challenging or eye-opening part of the task?
3. Skill Development (1 Mark) – How did this activity strengthen your technical or analytical thinking skills? How would you apply these skills in a future project or work situation?

Submission Format (Applicable only for Part A questions)

- File 1: Assignment Report (PDF, max 4 pages)
- File 2: Code notebook (.ipynb) . Mention the BITS ID and NAME of the group members who contributed to the assignment exercises.
- Screenshots/ Simulation output copy of the codes can be appended in the File 1 PDF report.

Tools: Python, Google Colab, PyBullet, or Gazebo

Deadline: As announced on eLearn



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Marking Scheme

The grading will be relative based on the class performance.

Key factors which will be considered for scoring includes (but not limited to): Creativity, Originality, Uniqueness, Relevance to the chosen domain's open problems, Code Modularity, Representation, Reproducibility, Similarity to AI tool solutions.

Component	Description	Marks
Part A – Application	Robot system, motion model, and workplace evaluation	7
Part B – Individual Reflection	Workplace connection, learning insights, and skill reflection	3
Total		10